

IMO 2026 — step225_run3

Problem 5

Let $\mathbb{R}_{>0}$ denote the positive real numbers. We determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ satisfying

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)} \quad (\forall x, y > 0). \quad (*)$$

1. A functional equation

Put $x = f(y)$ in $(*)$. From the left inequality we obtain

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y) \geq \frac{f(f(y)) + y}{2} \implies f(f(y)) \leq 2f(y) - y.$$

From the right inequality we obtain

$$\frac{f(f(y)) + y}{2} \geq \sqrt{f(y)f(y)} = f(y) \implies f(f(y)) \geq 2f(y) - y.$$

Hence equality holds for every $y > 0$:

$$\boxed{f(f(y)) = 2f(y) - y} \quad (\forall y > 0). \quad (1)$$

2. Differentiability of f

Fix an arbitrary $y_0 > 0$ and set

$$x = f(y_0) > 0.$$

For any t with $y_0 + t > 0$ and $|t|$ sufficiently small (so that $x + t/2 > 0$), apply the two inequalities of $(*)$ with $(x, y) = (x, y_0 + t)$.

Left inequality:

$$\sqrt{\frac{x^2 + f(y_0 + t)^2}{2}} \geq \frac{f(x) + y_0 + t}{2}.$$

Because $f(x) = f(f(y_0))$, by (1) we have $f(x) = 2f(y_0) - y_0 = 2x - y_0$. Thus

$$\frac{f(x) + y_0 + t}{2} = x + \frac{t}{2}.$$

Squaring (both sides are positive) gives

$$\frac{x^2 + f(y_0 + t)^2}{2} \geq \left(x + \frac{t}{2}\right)^2 \implies f(y_0 + t)^2 \geq x^2 + 2xt + \frac{t^2}{2}.$$

Since $f(y_0 + t) > 0$,

$$f(y_0 + t) \geq \sqrt{x^2 + 2xt + \frac{t^2}{2}}. \quad (2)$$

Right inequality:

$$\frac{f(x) + y_0 + t}{2} \geq \sqrt{x f(y_0 + t)} \implies x + \frac{t}{2} \geq \sqrt{x f(y_0 + t)}.$$

Squaring again yields

$$\left(x + \frac{t}{2}\right)^2 \geq x f(y_0 + t) \implies f(y_0 + t) \leq x + t + \frac{t^2}{4x}. \quad (3)$$

Now consider the difference quotient

$$\frac{f(y_0 + t) - x}{t}.$$

For $t > 0$ we have from (2) and (3)

$$\frac{\sqrt{x^2 + 2xt + \frac{t^2}{2}} - x}{t} \leq \frac{f(y_0 + t) - x}{t} \leq 1 + \frac{t}{4x}. \quad (4)$$

The function $h(t) = \sqrt{x^2 + 2xt + \frac{t^2}{2}}$ is differentiable at 0 with

$$h'(0) = \frac{2x}{2\sqrt{x^2}} = 1,$$

hence $\lim_{t \rightarrow 0^+} \frac{h(t) - x}{t} = 1$. By the squeeze theorem,

$$\lim_{t \rightarrow 0^+} \frac{f(y_0 + t) - x}{t} = 1.$$

For $t < 0$ we reverse the inequalities in (2) and (3) because division by a negative number flips the sign:

$$1 + \frac{t}{4x} \leq \frac{f(y_0 + t) - x}{t} \leq \frac{\sqrt{x^2 + 2xt + \frac{t^2}{2}} - x}{t}. \quad (5)$$

Again $\lim_{t \rightarrow 0^-} \frac{h(t) - x}{t} = 1$, so the squeeze theorem gives

$$\lim_{t \rightarrow 0^-} \frac{f(y_0 + t) - x}{t} = 1.$$

Consequently f is differentiable at y_0 and $f'(y_0) = 1$. Since y_0 was arbitrary,

$$f'(y) = 1 \quad (\forall y > 0).$$

Integrating, we obtain

$$f(y) = y + c \quad (\text{for some constant } c). \tag{6}$$

3. Determining the constant c

Because $f(y) > 0$ for every $y > 0$, (6) implies $y + c > 0$ for all $y > 0$. If $c < 0$ then for sufficiently small y we would have $y + c \leq 0$, contradicting positivity. Hence

$$c \geq 0.$$

4. Verification

Let $c \geq 0$ and set $f(x) = x + c$. For all $x, y > 0$ we check the original inequalities.

Left inequality:

$$\sqrt{\frac{x^2 + (y + c)^2}{2}} \geq \frac{x + y + c}{2} \iff 2x^2 + 2(y + c)^2 \geq (x + y + c)^2 \iff (x - y - c)^2 \geq 0,$$

which is true.

Right inequality:

$$\frac{x + y + c}{2} \geq \sqrt{x(y + c)} \iff (x + y + c)^2 \geq 4x(y + c) \iff (x - y - c)^2 \geq 0,$$

again true.

Thus every function of the form $f(x) = x + c$ with $c \geq 0$ satisfies (*).

5. Conclusion

All functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ satisfying the given conditions are exactly

$$\boxed{f(x) = x + c \quad \text{with } c \geq 0}.$$